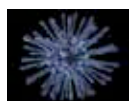




Scientists home in on FRBs

Researchers are investigating the extreme conditions that give rise to fast radio bursts.
<http://digit.in/fastrbs>



Space may be full of alien viruses

Viruses can survive in hazardous conditions but astrobiologists have not been looking for them so far.
<http://digit.in/alienviruses>

and extremely elongated interstellar asteroid" by Meech et al, in *Nature*. The object was at least 400 meters long, and was spinning rapidly, once every 7.3 hours. The surface had acquired a reddish hue from basking in the cosmic rays for millions of years. The dense asteroid was rocky or metallic. As the colour matches that of the Saturnian moon Iapetus, the composition of the asteroid is considered to be similar, a blend of minerals, carbon, iron and organic compounds. There is little or no water on 'Oumuamua. The visitor is similar to objects in the outer solar system, the trans neptunian objects or the remote and distant worlds in the Kuiper belt. The colour of 'Oumuamua suggested that it had characteristics similar to comets as well as the Trojan asteroids. While similar in composition to nearby celestial objects, the shape is nothing like anything in the Solar System. The most elongated known objects are only about three times as long as they are wide.

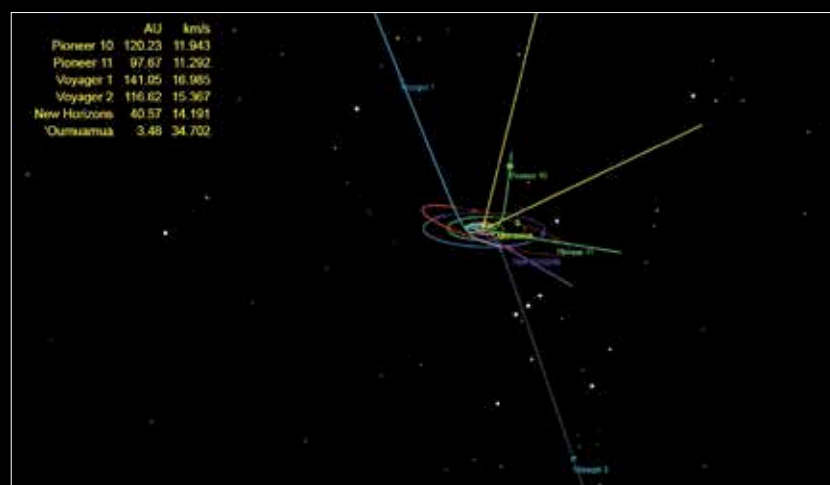
Another paper, published in the *The Astrophysical Journal Letters*, titled "Colors of the Interstellar Planetesimal 1I/'Oumuamua" by Bannister et al, does not entirely agree with the findings with the paper published by Meech et al. The significant differences are in the observed colours of the object, and how elongated the object is. According to the Bannister paper, the length of the axial ratio of the object is at least 5.3:1. Another paper titled "Time-resolved Color Photometry of Highly Elongated Interstellar Object 1I/'Oumuamua" by Bolin et al, published in the *Astrophysical Journal Letters* also found the colouration of 'Oumuamua consistent with Trojan asteroids in the solar system. This paper gave a range of between 3.5 to 10.3:1 for the axial ratio. The discrepancies in observations are because the window available for observing the object was very small.

The elongated nature of 'Oumuamua raised another question. An interstellar alien spaceship is most likely to have an elongated, stream-

lined shape, similar to a cigar, to reduce exposure to damage from interstellar dust. Breakthrough Listen and an initiative backed by Russian billionaire physicist Yuri Milner all scanned the object, to see if it was transmitting any signals. The scans were in line with the observations from the astronomers. There were no extraterrestrial signals found, and 'Oumuamua was confirmed to be a natural object. The elongated shape is believed to be because of a tidal distortion event before 'Oumuamua was ejected from the system that it was born in.

located within our Solar System. An interesting point raised by the paper is that astronomers on alien worlds studying the object in the distant future may trace 'Oumuamua back to our Solar System.

Our Solar System ejects comets and asteroids because of the interactions with the larger outer planets. Researchers believe that a similar mechanism leads to asteroids and comets being ejected from other systems as well. So far, the estimate for the frequency of such interstellar visits have been ridiculously low. Researchers now believe that



The trajectory and positions of Oumuamua along with our most distant space probes, as of February 2018.

Another paper, titled "'Oumuamua as a messenger from the Local Association" by Fabo Feng and Hugh RA Jones published in *The Astrophysical Journal Letters* used computers to simulate the motion of the asteroid. The study found 109 encounters with nearby stars, at a distance of less than 16 light years, or 5 parsecs. While the asteroid had a reddish hue, it was not very red, which indicates that 'Oumuamua is relatively young. The asteroid was traced back to a group of young stars in the Pleiades moving group, located in the constellation of Taurus. The cluster is visible to the naked eye. The asteroid was likely ejected from a binary pair of stars, from the warmer part of the system, where the asteroid belt is

between one to ten such objects pass through the Solar System every year. These have not been detected so far considering how faint they are, as well as how brief their visits last. Because of 'Oumuamua, there are plans to modify telescopes to make them more capable of detecting such interstellar objects in the future. Studying such objects will increase the understanding of our solar system, as well as the process of planet formation. Tracking hypervelocity asteroids is also important considering the serious destruction that can be caused by a collision with Earth. As the speeds of these interstellar objects are much greater than that of local asteroids, the devastation by an impact event would also be greater. 

Credit: orbisimulator.com